Multi-Task Self-Supervised Learning for Automated Measurement of Left Ventricular Ejection Fraction in Echocardiography

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Background & Motivation

Clinical Context: Left Ventricular Ejection Fraction (LVEF) is a key metric in cardiac function assessment.

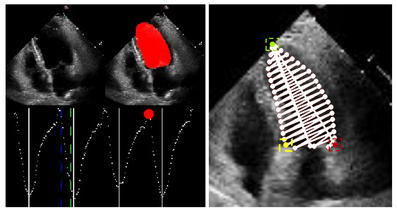


Figure 1. The process of clinicians processing echocardiograms. (a) Manual selection of key frames; (b) Annotation of the left ventricular endocardial border. The white dots represent points manually annotated by clinicians. The yellow box indicates the septal mitral annulus, the red box indicates the lateral mitral annulus, and the green box indicates the cardiac apex.

Challenge:

- Manual annotation is time-consuming and inconsistent.
- Existing deep learning methods fail to detect key anatomical landmarks (e.g., apex and mitral annulus) needed for Simpson's method.

Objective: Develop a novel lightweight, multi-task self-supervised framework that enables automated and accurate LVEF estimation under sparse annotation.

Ablation Experiment

The proposed **DSTS module** improves performance across various backbone networks (see Table 1). Moreover, combining **DSTS with self-supervised pretraining** further enhances model generalization with only minor computational overhead, as summarized in Table 2. These results confirm the practical value of our design for complex medical imaging tasks.

Table 1: Impact of DSTS on Different Backbones

Backbone	DSTS	Dice (%)↑	HD↓	OKS↑
MobileNet	✓	80.33	4.902	0.754
ResNet50	✓	92.05	3.791	0.742
SwinTrans	✓	90.94	4.083	0.728
PoolFormer	✓	89.44	5.216	0.731
DeiT	✓	86.62	5.428	0.717
EffFormer	✓	91.10	**3.645**	0.764
ReoViT	✓	91.89	3.812	0.790
MobileNetV3	✓	**92.47**	4.024	**0.795**

Table 2: DSTS and Pretraining Effects

Model	Dice↑	HD↓	OKS↑	Params	GFLOPs
Base	91.56	4.139	0.778	4.95M	0.23
Base+DSTS	92.47	4.024	0.795	5.01M	0.25
Base+Pre+DSTS	**93.54**	**3.512**	**0.803**	5.01M	0.25

References

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[2] Zeyun Zhao, Rong Wang, Jianzhe Gao, et al., "*Mask Matching Network for Self-supervised Few-shot Medical Image Segmentation*," in *Proceedings of the IEEE International Conference on Multimedia and Expo (ICME)*. IEEE, 2024, pp. 1–6.

[3] David Ouyang, Bryan He, Amirata Ghorbani, et al., "*Video-based AI for beat-to-beat assessment of cardiac function*," *Nature*, vol. 580, no. 7802, pp. 252–256, 2020.

Overview of SSP-MAEFNet

SSP-MAEFNet integrates self-supervised pretraining, weakly supervised learning, and multitask feature extraction into a unified framework, designed to enhance segmentation accuracy and landmark detection while ensuring clinical relevance.

Three-Stage Pipeline:

- **Stage 1:** Self-supervised pretraining via random masking and frame reconstruction. The model learns robust spatio-temporal features from unannotated echocardiographic videos.
- **Stage 2:** Weakly-supervised segmentation and landmark detection with a dual-branch decoder. A Deeply Separable Task-Specific (DSTS) module reduces feature entanglement.
- **Stage 3:** Ejection Fraction (LVEF) is calculated using the Simpson biplane method based on segmentation and landmark detection results.

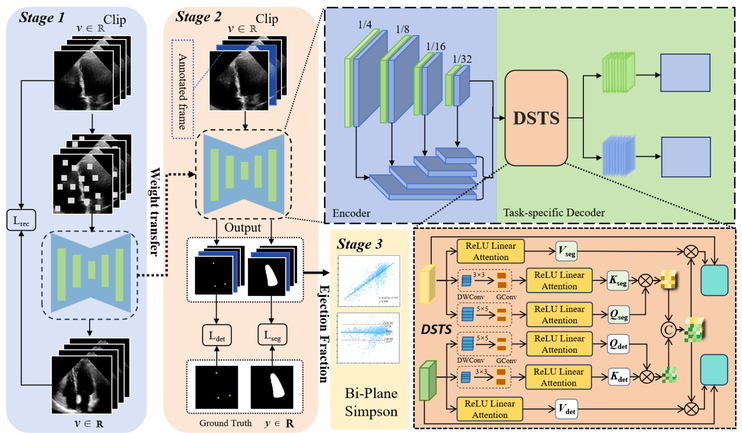


Figure 2. Overview of SSP-MAEFNet

This end-to-end design aligns with real-world clinical workflows, offering improved automation and reliability in echocardiogram analysis.

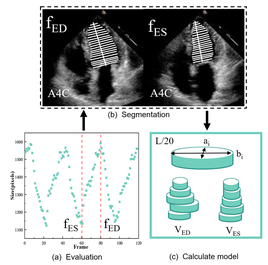


Figure 3. LVEF workflow. (a) Identify ED/ES frames; (b) Divide LV into disks; (c) Calculate LVEDV and LVESV.

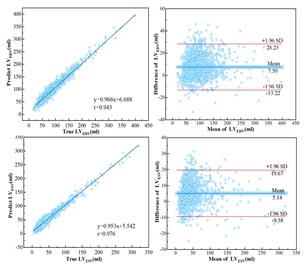


Figure 4. Distribution correlation and Bland-Altman plots diagram of predicted and true values of EDV, ESV, and LVEF.

Loss Functions

Stage 1: Self-Supervised Pre-Training

Optimized using only the reconstruction loss $\mathcal{L}_{\text{rec}}$ (mean squared error).

Stage 2: Fine-Tuning

Multi-task optimization with segmentation, landmark detection, and reconstruction.

- Reconstruction loss (MSE): $\mathcal{L}_{\text{rec}} = \frac{1}{N}\sum_{i=1}^{N}\left\|V^{(i)} - \hat{V}^{(i)}\right\|^2$
- Landmark detection loss: $\mathcal{L}_{\text{det}} = \frac{1}{2N}\sum_{i=1}^{N}\left[(x_{\text{pred}}^{(i)} - x_{\text{true}}^{(i)})^2 + (y_{\text{pred}}^{(i)} - y_{\text{true}}^{(i)})^2\right]$
- Segmentation loss (binary cross-entropy): $\mathcal{L}_{\text{seg}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right]$
- Total loss function: $\mathcal{L}_{\text{total}} = \lambda_{\text{rec}}\mathcal{L}_{\text{rec}} + \lambda_{\text{det}}\mathcal{L}_{\text{det}} + \lambda_{\text{seg}}\mathcal{L}_{\text{seg}}$

Results in EchoNet-Dynamic dataset

Table 1. Segmentation results

Model	DSC ↑	HD ↓	Params	FLOPs
U-Net	92.48	4.501	14.7M	10.4G
nnU-Net	92.83	4.379	19.2M	14.3G
EchoGraph	91.62	4.685	27.6M	21.5G
MemSAM	93.15	4.028	138.9M	109.6G
SegFormer	90.76	5.583	3.6M	11.5G
SepXception	92.88	4.475	55.8M	85.7G
TransBridge	91.02	5.213	**3.5M**	8.5G
DeepLabv3	90.17	5.429	19.1M	0.9G
simLVseg	93.32	3.795	125.8M	7.3G
EchoNet	92.23	4.253	39.6M	10.6G
EchoEFNet	92.29	4.643	32.4M	1.5G
Ours	**93.54**	**3.512**	5.0M	**0.25G**

Table 2. Landmark detection results

Model	OKS ↑	Params	FLOPs
HRNet	74.88	62.7M	3.7G
MSPN	76.39	27.5M	1.56G
TokenPose	76.70	5.2M	1.43G
AlphaPose	80.30	28.3M	3.51G
SimCC	76.00	25.1M	2.29G
RTMPose	80.90	28.6M	1.06G
Ours	**81.30**	**5.0M**	**0.25G**

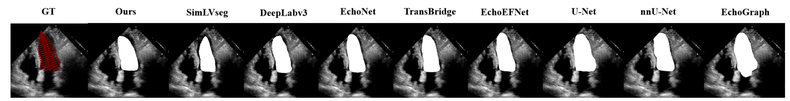


Figure 5. Segmentation results of different models.

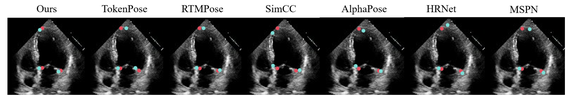


Figure 6. Landmark detection results.

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